

THEORIES OF THE POLICY PROCESS

FIFTH EDITION

Edited by **CHRISTOPHER M. WEIBLE**



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2 Punctuated Equilibrium Theory

Explaining Stability and Change in Public Policymaking

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Punctuated Equilibrium Theory (PET) seeks to explain a simple observation: although generally marked by stability and incrementalism, political processes occasionally produce large-scale departures from the past. Stasis, rather than crisis, typically characterizes most policy areas, but crises do occur. Large-scale changes in public policies are constantly occurring in one area or another of American politics as public understandings of existing problems evolve. Important governmental programs are sometimes altered dramatically, even if most of the time they continue as they have in previous years. Although both stability and change are important elements of the policy process, most policy models have been designed to explain—have been most successful at explaining—either the stability or the change. Punctuated Equilibrium Theory encompasses both.

In recent years, it has become clear that the general approach, developed in the early 1990s to explain US policymaking, applies to a broader set of governments than just the peculiar American system in which the theory was developed. Scholars around the world have confirmed aspects of the theory in a number of advanced democracies. In this chapter, we review Punctuated Equilibrium Theory, discuss new empirical studies in the United States and elsewhere, and interpret new theoretical developments. These developments have broadened PET to incorporate a general theory of disproportional information processing in the policy process. Over time, PET has changed not in the nature of its expectations but in the richness of its empirical support, especially cross-nationally, and in the development of more powerful foundations for its cognitive and institutional drivers.

In this chapter, we examine Punctuated Equilibrium Theory and its foundations in the longitudinal study of political institutions and in political decision-making (for other reviews, see John 2006b; Robinson 2006; Jones and Baumgartner 2012).¹ The theory has links to evolutionary biology (Eldridge and Gould 1972) though its application in the governmental context differs in important ways from its use in biology. Indeed, its intellectual roots are much closer to the study of complex systems (Érdi 2008), which

investigates complex interactions among component parts of a system, including political systems, that can generate unpredictability. Complexity in political systems implies that destabilizing events, the accumulation of unaddressed grievances, or other political processes can change the “normal” process of equilibrium and status quo on the basis of negative feedback (which dampens down action) into those rare periods when positive feedback (which reinforces action) leads to explosive change for a short while and the establishment of a new policy equilibrium.

We begin by discussing punctuated equilibrium in the context of the agenda-setting literature, extend the theory to national spending, and provide some recent evidence of punctuations and equilibria in US national government spending since World War II. Then we turn to how the theory has been generalized, including extensions to policymaking in US state and local governments as well as European national governments and in other governmental settings across the globe. These generalizations have been geographical (testing the ideas in new political systems), methodological (developing new statistical and qualitative means of testing the ideas), and substantive (expanding from only agenda setting and budgeting to a theory of institutional change). Next, we discuss in more detail how research on PET has developed since the first edition of this book. The chapter concludes with an assessment of the strengths and weaknesses of the PET approach to understanding public policymaking and notes the close linkage between the creation of a data infrastructure and the theoretical approach of analyzing policy dynamics.

Punctuated Equilibria in Public Policymaking

Since the pathbreaking work of E. E. Schattschneider (1960), theories of conflict expansion and agenda setting have stressed the difficulties disfavored groups and new ideas face in breaking through the established system of policymaking (Cobb and Elder 1983; Cobb and Ross 1997). The conservative nature of national political systems favors the status quo; multiple veto points, separation of powers, and other equilibrium-supporting factors have long been recognized. The key insight of PET is that, as a corollary of any system with a status quo bias, policy change will rarely be moderate: inertial forces for change are eliminated or kept to the smallest scale until and unless they are overpowered. The system generates a pattern of change characterized by stability most of the time, with dramatic shifts occurring when the inertial forces are overcome.

When Baumgartner and Jones (1993) analyzed a number of US policymaking cases over time and across a variety of issue areas, they found that (1) policymaking both makes leaps and undergoes periods of near stasis as issues emerge on and recede from the public agenda, (2) American political institutions exacerbate this tendency toward punctuated equilibria, and (3) policy images play a critical role in expanding issues beyond the control of the specialists and special interests that occupy what they termed “policy monopolies.”

Baumgartner and Jones (1991, 1993) saw that the separated institutions, overlapping jurisdictions, and relatively open access to mobilizations in the United States combine to create a dynamic between the politics of subsystems and the macropolitics of Congress and the presidency—a dynamic that usually works against any impetus for change but occasionally reinforces it. For example, mobilizations are often required to overcome entrenched interests, but once under way they sometimes engender large-scale changes in policy. The reason is that once a mobilization is under way, the diffuse jurisdictional boundaries that separate the various overlapping institutions of government can allow many governmental actors to become involved in a new policy area. Typically, the newcomers are proponents of changes in the status quo, and they often overwhelm the previously controlling powers. Institutional separation often works to reinforce conservatism, but it sometimes works to wash away existing policy subsystems.

In short, American political institutions were conservatively designed to resist many efforts at change and thus to make mobilizations necessary to overcoming established interests. The result has been institutionally reinforced stability interrupted by bursts of change. These bursts have kept the US government from becoming a gridlocked Leviathan despite its growth in size and complexity since World War II.

However, increasing partisan and affective (or emotional) polarization have led to greater friction in the system, and hence more difficulties in breaking through the extended periods of gridlock (Theriault 2008; Thurber and Yoshinaka 2015; Hetherington and Weiler 2018). This has led to longer periods in which there is an imbalance between the changing nature of problems facing the American system and the potential solutions available for addressing these problems. The longer the period of stability in which problems are allowed to fester because of the friction in the institutional mechanisms and the partisan polarization in politics, the larger the policy punctuations must be to re-calibrate the connection between problems and solutions. This longer period of stasis in which inaction dominates adaptive problem-solving has become a major feature of modern political systems. It leads to larger punctuations and the potential that the policy change overreacts to the problem, leading to what we term a policy bubble. We discuss this heightened friction feature later in this chapter.

The American political system has developed in a way that partially overcomes this dilemma through bureaucratic rule-making. Other political systems have experienced similar stresses and have responded in similar ways. The system of rule-making overseen by congressional committees with expertise in the area has become a complex, interactive system because of overlap between policy areas. This has led to the potential for under-the-radar adaptive problem-solving that avoids the gridlock of the major policy-making branches.

Redford (1969) differentiated between subsystem politics and macropolitics. Baumgartner and Jones extended Redford's insight and combined it with the issue expansion and contraction insights of Schattschneider (1960) and

Downs (1972) to form a theory of long-term agenda change and policy-making. Thus, at the core, the literature on agenda setting has always been concerned with the power of specialized communities of experts and the degree to which they operate with relative autonomy from the larger political system or are subject to more intense scrutiny. Because the members of any professional community of experts (say, farmers, nuclear engineers, or members of the military) may prefer more spending on “their” policy, political scientists have long been concerned with tracing the relative power of these shared interest communities.

No political system features continuous discussion on all issues that confront it. Rather, discussions of political issues are usually disaggregated into a number of issue-oriented policy subsystems. These subsystems can be dominated by a single interest, can undergo competition among several interests, can disintegrate over time, or can build up their independence from others (Sabatier 1987; Worsham 1998). They may be called “iron triangles,” “issue niches,” “policy subsystems,” or “issue networks,” but any such characterization can be considered only a snapshot of a dynamic process (Baumgartner and Jones 1993, 6). Whatever name one gives to these communities of specialists operating out of the political spotlight, most issues, most of the time, are treated within such a community. Nonetheless, within the spotlight of macropolitics, some issues catch fire, dominate the agenda, and result in changes in one or more subsystems. The explanation for the same political institutions producing both stasis and punctuations can be found in the processes of agenda setting—especially the dynamics produced by bounded rationality and serial information processing. These affect the interactions between communities of experts and the larger political system.

Serial and Parallel Processing

Herbert Simon (1957, 1977, 1983, 1985) developed the notion of bounded rationality to explain how human organizations, including those in business and government, operate. He distinguished between parallel and serial processing. Individuals devote conscious attention to one thing at a time, so decision-making must be done in serial fashion, one thing after the other. Organizations are somewhat more flexible. Some decision structures are capable of handling many issues simultaneously, in parallel. Political systems, like humans, cannot simultaneously consider all the issues that face them at the highest level, so policy subsystems can be viewed as mechanisms that allow the political system to engage in parallel processing (Jones 1994). Thousands of issues may be considered simultaneously in parallel within their respective communities of experts.

Sometimes parallel processing within distinct policy communities breaks down, and issues must be handled serially. In the United States, the macropolitical institutions of Congress and the public presidency engage in serial processing, whereby high-profile issues are considered, contested, and decided one—or at most a few—at a time. An issue moves higher on the

political agenda usually because new participants have become interested in the debate: "When a policy shifts to the macro-political institutions for serial processing, it generally does so in an environment of changing issue definitions and heightened attentiveness by the media and broader publics" (Jones 1994, 185). Issues cannot forever be considered within the confines of a policy subsystem; occasionally macropolitical forces intervene as friction or gridlock is overcome. The intersection of the parallel-processing capabilities of the policy subsystems and the serial-processing needs of the macropolitical system creates the non-incremental dynamics of lurching that we often observe in many policy areas. Agenda access does not guarantee major change, however, because reform is often blunted in the decision-making stage (Wolfe 2012). But this access is a precondition for major policy punctuations.

An interesting but largely untested area is the likelihood of substantial policy change in the absence of salience or agenda access. This could come, for example, by shifting norms within a professional community, but without broad social discussion, or by the accumulation of many small changes each moving in the same direction. Such slow accretion of policy changes moving in the same direction can correct maladaptive policies, and many have been documented (see, e.g., Jacob 1988 on the issue of US divorce law or Armstrong 2003 on the development of the US medical consensus on the harmful effects of any alcohol during pregnancy).

However, problems can aggregate in a similar manner, and such problem aggregation is easier for a policymaking system to ignore than the emergence of a crisis. Epp and Thomas (2023) find that such patterns lead to worse problems and more intense policy punctuations.

When dominated by a single interest, a subsystem is best thought of as a policy monopoly. A policy monopoly has a definable institutional structure responsible for policymaking in an issue area, and its responsibility is supported by some powerful idea or image. This image is generally connected to core political values and can be communicated simply and directly to the public (Baumgartner and Jones 1993, 5–7). Because a successful policy monopoly systematically dampens pressures for change, we say that it contains a negative feedback process. Yet policy monopolies are not invulnerable forever.

A long-term view of US policymaking reveals that policy monopolies can be constructed and can collapse. Their condition has an important effect on policymaking within their issue areas. If the citizens excluded from a monopoly remain apathetic, the institutional arrangement usually remains constant, and policy is likely to change only slowly (the negative feedback process). As pressure for change builds up, it may be resisted successfully for a time. But if pressures are sufficient, they may lead to a massive intervention by previously uninvolved political actors and governmental institutions. Generally, this requires a substantial change in the supporting policy image. As the issue is redefined or previously dormant dimensions of the debate become salient, new actors that had previously stayed away feel qualified to exert their authority. These new actors may insist on rewriting the rules and

changing the balance of power, which will be reinforced by new institutional structures as previously dominant agencies and institutions are forced to share their power with groups or agencies that gain new legitimacy.

Thus, the changes that occur as a policy monopoly is broken up may be locked in for the future as institutional reforms are put in place. These new institutions remain in place after public and political involvements recede, often establishing a new equilibrium in the policy area that lasts well after the issue backs off the agenda and into the parallel processing of a (newly altered) policy community. Important elements of this process are the power, prestige, and legitimacy of the previously established policy monopoly. Such “incumbents” seek to maintain their control. Whether they are or are not discredited enough by policy failures to lose their influence depends on both their levels of policy success and prestige and the strength of those who seek to replace them (see Baumgartner 2013).

Friction in the policymaking system and associated policy monopolies can be disturbed by either exogenous (i.e., external) events or endogenous (internal) forces. Oftentimes, a combination of both occurs, putting intense pressure on the governing system. The COVID pandemic offers the potential of comparing the responses of different systems with different degrees of friction to the challenges generated by the pandemic.

In the United States, public health officials first responded slowly to the challenge, failing to sponsor enough testing and relying on poor-quality but locally developed tests. These proved to be limited obstacles, and as time went on the system improved its performance.

China, an authoritarian system with more friction and less responsiveness to information resisted by central authorities, at first responded quickly and with great efficiency due to the attention directed to the problem by authorities. Authorities quickly developed a locally developed vaccine, but the vaccine proved to be less effective than other non-Chinese versions. Yet, authorities held to the early strategy of local vaccines and complete lockdowns of outbreaks, which continued to buffer the economy.

High levels of friction do not mean that no action is possible; it means that it is difficult to absorb new information into the decision-making system because the set of rules makes that difficult. It does mean that once set courses of action are hard to shift. As a consequence, where friction is very high, adjustment can be quick, but it is not adaptable over the long run, leading to larger policy punctuations when they cannot be avoided. In lower friction systems in which information is continually processed, adjustment is smoother, and the resulting punctuations are less severe. Only a system with perfect proportionality between incoming signals and immediate governmental response would avoid punctuation, but all complex decision-making systems are subject to the types of information overload that generate the disproportionate response characteristic of PET. Those with higher levels of friction have more severe punctuations, however. How institutions process information is key in comparing high and low friction systems, and we return to this distinction later in this chapter.

Positive and Negative Feedback

Punctuated Equilibrium Theory includes periods of equilibrium or near stasis, when an issue is captured by a subsystem, and periods of disequilibrium, when an issue is forced onto the macropolitical agenda. When an issue area is on the macropolitical agenda, small changes in objective circumstances can cause large changes in policy, and we say that the system is undergoing a positive feedback process (Baumgartner and Jones 2002). Positive feedback occurs when a change, sometimes a fairly modest one, causes future changes to be amplified. Observers often use terms like “feeding frenzy,” “cascade,” “tipping point,” “momentum,” or “bandwagon effect” to characterize such processes. Negative feedback, however, maintains stability in a system, somewhat like a thermostat maintains constant temperature in a room.

Note that our use of the phrases positive and negative feedback differs from that of some other scholars in different contexts. For example, sometimes “negative feedback” is understood to mean criticism, and “positive feedback” means praise. To avoid confusion because of these similarities in the language used to describe very different things, we recommend caution. Our use of “positive” and “negative” is mathematical in origin. When faced with an incoming signal, a negative feedback system responds in the opposite way as the signal, as a thermostat puts out cold air in the face of warming ambient temperatures. In a “positive feedback” system, the response is in the same direction as the input, not in the opposite direction. If this is a closed loop, a positive feedback system can be explosive. Thus, readers should use caution in their use of terms and understand the underlying meaning of positive and negative feedback *systems*. These differ from more colloquial uses of the terms positive and negative feedback.

Physical scientists have studied large interactive systems that are characterized by positive feedback. Physical phenomena like earthquakes can result from fairly modest changes. Pressure inside the earth builds up over time and eventually causes the tectonic plates on the planet’s surface to shift violently during an earthquake. Similarly, if we drop grains of sand slowly and constantly onto a small pile of sand in a laboratory, most of the time the pile remains in stasis, with occasional landslides, some of which are minor, and others of which are huge (Bak 1997). A landslide may not be caused by a large-scale event; it may be caused by the slow and steady buildup of tiny changes. Like earthquakes and landslides, policy punctuations can be precipitated by a mighty blow, an event that simply cannot be ignored, or by relatively minor events that accumulate over longer periods. What determines whether an issue catches fire with positive feedback? The interaction of changing images and venues of public policies.

As an example of positive feedback in policymaking, let us take the case of the involvement of the US national government in criminal justice. Before the late 1960s, federal involvement in crime policy was relatively modest. At the end of that decade, however, the Lyndon Johnson administration initiated several new federal grant-in-aid programs to assist state and local

governments in crime prevention and control. Congress passed the Omnibus Crime Control and Safe Streets Act in 1968; between 1969 and 1972 federal spending on crime and justice doubled in real dollar terms.

What happened? Crime was rising during this period, but more importantly other trends highlighted the increasing insecurity citizens were feeling, causing people and government officials to direct their attention to the crime problem. Three important measures of attention and agenda access came into focus all at once: press coverage of crime stories, the proportion of Americans saying that crime was the most important problem (MIP) facing the nation, and congressional hearings on crime and justice. All of this happened as major urban disorders swept many American cities. In the words of John Kingdon, a window of opportunity had opened, and federal crime policy changed in a major way. After 1968, the three trends fell out of focus, going their own ways, and crime policy moved back into the subsystem arena. It is not possible to say which of the three variables was primary; all three were intertwined in a complex positive feedback process. In a classic pattern, public attention to crime jumped, press coverage focused on the problem, and Congress scheduled hearings. The issue left its normal subsystem home, with incremental adjustments, and entered the realm of macropolitics. Congress passed a major law, and spending increased in a major punctuation. US crime policy at the federal level is still powerfully affected by decisions that were made during this surge of attention on the “war on crime” and those that later reinforced them.

Jones, Thomas, and Wolfe (2014) define a policy bubble as sustained overinvestment in a policy solution (or instrument) or set of solutions relative to the efficiency of the policy solution in achieving goals. To illustrate they study three potential policy bubbles: crime control, privatization and contracting, and charter schools and vouchers. They conclude that the first two policies clearly generated overinvestment bubbles, but the third did not, primarily because countermobilization by affected interests limited the positive feedback effects.

Baumgartner and colleagues (2021) give another example: how US politicians at all levels promoted extremely harsh criminal justice penalties, including life without the possibility of parole, in the 1990s and how the obvious mathematical consequence of this, exploding numbers of geriatric prisoners needing expensive medical care but presenting little threat to community safety, were ignored. Over-reactions such as this are not uncommon in policy. One major reason for this is the power of ideas, or “policy images,” and how these change.

Policy Images

Policymakers, political activists, and citizens at large develop images of policies that interpret both the problem and the policy solution being discussed. Policy images are a mixture of empirical information and emotive appeals.

Such images are, in effect, information—grist for the policymaking process. The factual content of any policy or program can have many different aspects and can affect different people in different ways. When the policy image is connected to what we might call a story-line, the result is a policy narrative (see Chapter 5 in this volume).

When a single image is widely accepted and generally supportive of the policy, it is usually associated with a successful policy monopoly. When there is disagreement over the proper way to describe or understand a policy, proponents may focus on one set of images while their opponents refer to a different set. For example, when the image of civilian nuclear power was associated with economic progress and technical expertise, its policymaking typified a policy monopoly. When opponents raised images of danger and environmental degradation, the nuclear policy monopoly began to collapse (Baumgartner and Jones 1993, 25–28, 59–82).

A new image may attract new participants, and the multiple venues in the American political system constitute multiple opportunities for policy entrepreneurs to advance their cases. Federalism, separation of powers, and jurisdictional overlaps not only inhibit major changes during periods of negative feedback but also mean that a mobilization stymied in one venue may succeed in another. The states can sometimes act on a problem that has not advanced onto the national agenda, and vice versa. The US system of multiple policy venues is an important part of the process of disrupting policy monopolies during periods of positive feedback. Today the difference in policy images between states governed by Republicans and those governed by Democrats is striking on many issues, including the acceptance of vaccines, firearms policies, abortion regulation, and voting laws. Many of these policies have experienced the politics of positive feedback as states rush to copy innovations in other states, but are often stopped at the red (Republican)–blue (Democratic) boundary.

In summary, subsystem politics is the politics of equilibrium—the politics of the policy monopoly, incrementalism, a widely accepted supportive image, and negative feedback. Subsystem decision-making is decentralized to the iron triangles and issue networks of specialists in the bureaucracy, legislative subgroups, and interested parties. Inertia tends to dampen departures from established interests until political mobilization, advancement on the governmental agenda, and positive feedback occur. At that point, issues spill over into the macropolitical system, making possible major change.

Macropolitics is the politics of punctuation—the politics of large-scale change, competing policy images, political manipulation, and positive feedback. Positive feedback exacerbates impulses for change; it overcomes inertia and produces explosions or implosions from former states (Baumgartner and Jones 1993; Jones, Baumgartner, and Talbert 1993; Jones 1994; Talbert, Jones, and Baumgartner 1995).

Rigorous qualitative and quantitative studies again and again find strong evidence of the process, including studies on regulatory drug review (Ceccoli 2003), environmental policy (Repetto 2006), education (Manna

2006; Robinson 2004), firearms control (True and Utter 2002), and regulation of state hospital rates (McDonough 1998).

This sweeping depiction of issue dynamics may hide a great deal of variability in the operation of policy subsystems. For example, Worsham (1998) examines three different subsystem types and finds substantial variation in the actors' ability to control attempts to shift conflict from the subsystem level to the macropolitical level by appealing to Congress. In his study of federal land management, Wood (2006) shows that even conflictual subsystems can sometimes avoid disruption through conflict-management strategies. More generally, this suggests that institutional arrangements can affect the magnitude of punctuations—a point to which we return later in this chapter.

Boundedly Rational Foundations and the Centrality of Decision-Making

Embedded in the Punctuated Equilibrium Theory of policy change is an implicit theory of individual and collective decision-making. Large-scale punctuations in policy spring from a change in either preferences or attentiveness to different aspects of preferences. If we regard preferences as relatively stable, and political actors do not change, then how can we explain major changes in government policy?

Part of the answer is found in Jones's (1994) analysis of serial attention, and part is found in Jones and Baumgartner's (2005) analysis of the disproportionate nature of human individual and collective information processing. They expand on these themes in *The Politics of Information* (Baumgartner and Jones 2015). Jones (1994) argued that individual and collective decision changes, including choice reversals, do not spring from rapid flip-flops of preferences or from basic irrationality (choosing to go against our own preferences); they spring from shifts in attention. In individuals, serial attentiveness means that attention is given serially to one thing, or at most a few things, at a time (Simon 1977, 1983). Although reality may be complex, changing, and multifaceted, individuals cannot smoothly integrate competing concerns and perspectives. We usually focus on one primary aspect of the choice situation at a time (Simon 1957, 1985; Jones 1994). A shift in the focus of attention of a group can lead to a disjointed change in preferred alternatives, even when the alternatives don't change (Jones 1994).

More generally, bounded rationality undergirds all policy change because the mechanisms associated with human cognitive architecture are also the characteristic of organizations, including governments (Jones 2001, 2003; Kahneman 2011; Jones and Thomas 2012). Bounded rationality is the decision-making underpinning of both the punctuated equilibrium and the advocacy coalition approaches (see Chapter 4 in this volume), but the theories emphasize different aspects of the process. Punctuated equilibrium is based on the serial processing of information and the consequent attention shifts, whereas the advocacy coalition approach traces policy dynamics to the belief systems of coalition participants (Leach and Sabatier 2005).

Bounded rationality was wedded early to incrementalism, minor adjustments to the status quo (Lindblom 1959; Wildavsky 1964). Yet incrementalism proved to be an incomplete explanation of government policymaking, because budget changes were sometimes very large. And the major non-incremental changes could not be integrated with the incremental changes in a single model.

With Jones's (2001) reconceptualization, however, boundedly rational decision-making is a foundation for both major and minor changes—for both punctuations and equilibria. In the case of public policymaking, the twin foundations of conservative and overlapping political institutions and boundedly rational decision-making (especially the role of images in dampening or exacerbating mobilizations against entrenched interests) combine to create a system that is both inherently conservative and liable to occasional radical change.

Information Processing

Punctuated Equilibrium Theory is at base a theory of organizational information processing. Governments are complex organizations that act on the flow of information in producing public policies (Jones, Workman, and Jochim 2009). The manner in which public policy adjusts to these information flows determines the extent of bursts of activity in the system. The general punctuation hypothesis suggests that information processing is disproportionate. That is, policymaking alternates between periods of under-reaction and over-reaction to the flow of information coming into the system from the environment (Jones and Baumgartner 2005). This reaction may stem from a vivid event that symbolizes everything that is wrong (Birkland 1997) or from the accumulation of problems over longer periods. In either case, how the policymaking system allocates attention to the problem is a critical component of problem recognition and subsequent policy action, but so are the institutional arrangements responsible for policymaking.

One would expect a policymaking system, then, to be more subject to punctuations when it is less able to adjust to the changing circumstances it faces. Jones and Baumgartner (2005) show that a perfect pattern of adjustment to a complex, multifaceted environment in which multiple informational input flows are processed by a political system will yield a normal distribution of output changes. As a consequence, the extent of the adjustment of a policy system may be gauged by a comparison of its distribution of policy outputs with the normal curve. In an important sense, the more normally public policy changes are distributed, the better the policymaking system is performing (in the sense of producing efficient adjustment to environmental demands).

Adler and Wilkerson (2012) have developed what amounts to a new theoretical approach to the study of congressional behavior by adopting an information and problem-solving approach. They note, for example, that much of the US congressional workload is organized around a small number

of “must-pass” pieces of legislation and that lawmakers structure things to ensure that bills that “must” be passed will periodically arise. That is, Congress plans policy punctuations. This dynamic has become more severe as partisan polarization has become more severe, leading to packaging of bills into a “reconciliation” package designed to avoid the friction-producing Senate filibuster.

Complex interactions, however, cannot always be confined to activity within fixed institutional frameworks. The entire policymaking system can evolve, and the pieces of the system can feed back into the whole, actually changing the decision-making structures that acted as policy venues in the first place. Richardson (2000) argues that this is happening in European policymaking, and Daviter (2009) reinforced this point in the EU context as well. In the United States, a technical budget mechanism called reconciliation has become the major mechanism for policy change, because it is not subject to another arbitrary rule, the filibuster. The policy process can be viewed as a complex, evolving system with the decision-making structure itself subject to change. Such changes can add friction to a policymaking system, as in the case of the evolution of the filibuster to a more powerful mechanism for blocking policies.

The Politics of Information and the Pathology of Punctuations

The concept of punctuated equilibria in policy studies is based on the theory of policy subsystems, in particular on ideas originating with E. E. Schattschneider and Emmette Redford (see above). In the early work on punctuated equilibria, major policy changes were seen as natural outcomes of normal democratic processes. Friction in policymaking systems was a natural outcome of parallel processing in policy subsystems, rules that limit policy action, and the cognitive capacities of human actors that limit information-processing abilities. As a consequence, changes in collective attention were necessary to overcome the bias of the status quo, leading to disjointed large-scale policy changes. As the *Politics of Attention* showed, such changes could happen even in the absence of crises in the policymaking environment.

It has become clear in recent years that this view is incomplete. Political systems may be designed with such a high level of friction that they strongly resist change, so when major changes come (and they will), the changes can be highly destructive. The friction dynamic implies that the more centralized and authoritarian the regime, the larger the policy punctuations will be because the system is less able to adjust to the flows of information from the environment. Lam and Chan (2014) show that policy changes were greater when Hong Kong was more centralized but abated as the political system democratized. Chan and Zhao (2016) develop what they call the “information disadvantage of authoritarianism.” They show that large policy punctuations occur in authoritarian China relative to democratic regimes and that punctuations are larger in regions of China with less social

discontent. Given the lack of other input means, discontent is one of the few mechanisms for stressing problems that exist. It is a mechanism of information processing in closed systems but can be important in democracies, especially those with high friction, as well. For example, more centralized agency structures within government may lead to patterns of less stable outputs (May, Workman, and Jones 2008). Epp (2018) shows that firms in decentralized markets are less punctuated than firms in less competitive situations.

Punctuations are inevitable, but their size and distribution are not. How can policymaking systems be designed so that the size of punctuations can be minimized? It is well established in the policy process literature that the dynamics affecting the discovery and interpretation of policy problems are distinct from the search for solutions. In *Human Problem-Solving*, Newell and Simon (1972) found that people solving problems tended to return to prepackaged sets of policy solutions when encountering a superficially similar problem to one they had solved before (see Jones 2001 for a discussion). The garbage can theory of Cohen, March, and Olsen (1972), extended by Kingdon (1984) to what is now called the Multiple Streams Framework (Herweg, Zahariadis, and Zohlnhöfer 2017), treats problem dynamics and solution search as separate processes at the systems level.

Baumgartner and Jones's *Politics of Information* develops the thesis that problem discovery and definition require a different organizational system than solution search. They show that policymaking systems may reach sub-optimal equilibria by suppressing attributes in a complex problem space. They develop the thesis that "entropic search," in which multiple potentially competing jurisdictions of government agencies and legislative committees are considered, yields a superior (in the sense of more consistent input streams of information) depiction of the problem space. But often collaborations among experts are better at designing solutions than a cacophony of competing voices is. Organizational design may need to differ for detecting and defining problems and designing solutions.

Although punctuations are unavoidable, better governance systems tend to minimize the disruption from such punctuations. Crises, of course, can be unexpected, what Taleb (2007) calls black swans when they are particularly extreme. But many and probably most crises are foreseeable to some extent; open and even confusing policy systems are better equipped to detect such potential crises than are more centralized and less adaptive ones. In a study of budgeting outcomes in a range of authoritarian and democratic political systems, Baumgartner and colleagues (2017) found that authoritarian systems had greater punctuations. Apparently, the institutional power of the authoritarian leader was outweighed by the informational disadvantage of that form of government.

The Dual Role of Friction in Policymaking

Punctuated equilibrium in policy studies applies to a particular situation: when political conflict is expanded beyond the confines of expert-dominated

policy subsystems to other policymaking venues. It relies on the mechanism of policy image—the manner in which a policy is characterized or understood—and a system of partially independent institutional venues within which policy can be made. The general punctuation hypothesis generalizes this basic framework to situations in which information flows into a policymaking system, and the system, acting on these signals from its environment, attends to the problem and acts to alleviate it if necessary (Jones, Sulkin, and Larsen 2003; Jones and Baumgartner 2005).

This translation is not smooth, however, because decision-making activities are subject to decision and transaction costs. These are costs that policymakers incur in the very process of making a decision. Participants in a policymaking system must overcome these costs to respond to the signals from the environment, which themselves are uncertain and ambiguous. There are two major sources of costs in translating inputs into policy outputs. The first consists of cognitive costs: political actors must recognize the signal, devote attention to it, frame the problem, and devise solutions for it. Cognitive costs also involve the resistance that emerges from ideological rigidity. The second source consists of institutional costs: the rules for making policy generally act to maintain stability and incrementalism. Some of these rules are higher in friction than others. In the case of US national institutions, constitutional requirements of supermajorities to pass legislation mean that policy outputs will be more punctuated than the information coming into government.

Policymaking institutions add friction to the process of translating inputs into policy outputs. This friction acts to delay action on issues until enough pressure develops to overcome this institutional resistance. Then there is a lurch or punctuation in policymaking. This framework has proved useful in understanding differences among political systems, which, after all, add friction to the policymaking process in different ways. Some social movement theorists have critiqued policy process approaches as too narrow, but they do stress issue dynamics (Kenny 2003). A more general formulation may lead to grappling with how one might integrate the voluminous work on social movements with punctuated change within institutional frameworks. (For a discussion of how social movements can be integral to the process of institutional change, see Jones, Theriault, and Whyman 2019.)

Recently, E. J. Fagan (2022) has developed an approach that integrates under-reaction to policy problems and over-reaction to them (which in the extreme he terms “policy disasters”). Most policy disasters are avoidable, resulting from major mistakes by government. High friction systems are subject both to more punctuations due to under-reaction and to those associated with over-reactions.

Punctuations and Stability in Government Spending

Over the past twenty-five years, Punctuated Equilibrium Theory has been extended to produce an agenda-based model of governmental budgeting (Jones, Baumgartner, and True 1998; True 2000; Jones and Baumgartner

2005; Jones et al. 2009; see also Jensen, Mortensen, and Serritzlew 2016, 2019). Collectively, government decision makers usually process information in a parallel way through subsystems, policy monopolies, iron triangles, and issue networks. When that happens, budgets change only incrementally.

However, sometimes issues move from subsystem politics to macropolitics, and national attention in Congress and in the presidency is, of necessity, focused on one or a few high-profile items at a time. In the attention limelight of the macropolitical institutions, policies and programs can make radical departures from the past, and budgets can lurch toward large changes. The study of budgets grew from a desire to construct a comprehensive test of PET; Jones and Baumgartner (2005) presented data on thousands of budget changes at the federal level, demonstrating a pattern in the overall distribution that was consistent with the theory. Since then, an entire theory of budgeting has developed based on bounded rationality (Jones et al. 2009).

Choice situations are multifaceted; yet decision makers tend to understand choices in terms of a circumscribed set of attributes, and they tend to have considerable difficulty making trade-offs among these attributes. If a given policy promotes economic growth but simultaneously has some negative consequences in terms of human rights, one or the other of those competing values may be at the forefront of decision makers' attention. If attentiveness to these two dimensions was to shift—say, as a result of scandal or changes in the composition of the group of decision makers, as sometimes occurs—then the chosen policy might shift dramatically as well. In general terms, Jones (2001) has noted that decision makers tend to stick with a particular decision design (a term that refers to the attributes used in structuring a choice) until forced to reevaluate it.

Government spending reacts to both endogenous and exogenous forces. Such influences may include changing levels of public attention, striking and compelling new information, or turnover in the composition of the decision-making body through elections. When changing external circumstances force us out of an old decision design, the result is often not a modest adjustment but a major change. Yet subsystem politics and the bureaucratic regularity of annual budget submissions constitute endogenous forces that favor continuing with the same decision design. As a consequence, budget decisions tend to be either static, arrived at by applying the current decision design and subsystem institutions to the new choice situation, or disjointed, arrived at by utilizing a different decision design and macropolitical institutions that may incorporate new attributes into the choice structure or shift attention from one dimension to another.

This view of the budget process leads us to expect that annual budget changes within a given spending category will not be distributed in the normal, bell-shaped curve. Rather, these changes should reflect the non-normal distributions found in earthquakes and other large interactive systems (see Mandelbrot 1963; Padgett 1980). The “earthquake” budget model anticipates many minuscule changes, few moderate changes, and many large changes (True 2000).

If budgets generally change very little, but occasionally change a great deal, annual budget changes will be distributed leptokurtically. Their univariate distribution should have a large, slender central peak (representing a stability logic), weak shoulders (representing difficulty in making moderate changes), and big tails (representing episodic punctuations). A normal, or Gaussian, distribution of budget changes would be found if continuous dynamic adjustment were the primary decision mechanism (Davis et al. 1966; Padgett 1980).

The model implies that punctuations ought to occur at all levels of policymaking and at all levels of the budget and should not be driven simply by external (exogenous) factors in a top-down manner. This is a consequence of two factors. First, budget decisions are hostage to the statics and dynamics of selective attention to the underlying attributes structuring a political situation. Second, the theory of punctuated policy equilibrium is based in part on a bottom-up process in which policy change may occur in isolated subsystems, may spill over into other, related subsystems, or may be affected by exogenous shocks (Jones, Baumgartner, and True 1998).

Punctuations in Budget Theories

The empirical and theoretical literature on public budgeting provides ample precedent to expect budget punctuations, beginning as shown above with Davis, Dempster, and Wildavsky (1966). This study focused on the use by decision makers of budget decision rules. These rules, understood by participants and offering a stable organizational environment for decision-making, were based on the concepts of base and fair share, which led to incrementalism in both process and output. But Davis, Dempster, and Wildavsky (1974, 427) later added that “although it is basically incremental, the budget process does respond to the needs of the economy and society, but only after sufficient pressure has built up to cause abrupt changes precipitated by these events.”

The policymaking literature is replete with models of exogenously forced policy change (Krasner 1984). But external forces may sometimes damp down policy changes, not always cause punctuations. Scholars studying public representation see changes in public policy as exogenously driven by changes in public opinion (Stimson, MacKuen, and Erikson 1995) in a thermostat-like process (Soroka and Wlezien 2010). The thermostatic model adds to our understanding of why budget changes often vary within a limited range. If friction is incorporated, then one would predict smoother connections between changes in public opinion and changes in public policy where institutional friction is lower. Lower friction systems imply better representation of public opinion (Fagan, Jones, and Wlezien 2017). The key is the interaction between institutional friction and external changes.

The agenda-based budget model assumes that budgeting is a stochastic process. This just means that we study distributions of budget changes rather than examine them separately. It remains extremely difficult (and perhaps impossible) to specify precise causal linkages among all of the variables that

interact in complex ways to produce changes in all of the line items of annual national budgets. The agenda model predicts a distribution that has few moderate changes but a relatively large number of very large changes, because different forces act on the different policies that comprise a government spending pattern (Jones, Baumgartner, and True 1998; Jones and Baumgartner 2005; Jones et al. 2009; Jones, Zálányi, and Érdi 2014).

The Distribution of Budget Changes

We first presented tests of this hypothesis in the first edition of this book; since then, policy process scholars have produced a virtual explosion of work on the distribution of budget changes. Figure 2.1 shows a frequency distribution of year-to-year percentage changes of ten budget categories across twenty-four OECD countries for the period 1996–2011. Each observation is an annual percentage change for a country for a particular budget category, and the dashed line represents a normal distribution with a similar mean and variance as the observed distribution. A further explanation of the data can be found in Fagan, Jones, and Wlezien (2017).

The distribution is clearly leptokurtic and positively skewed. Note the very strong central peak, indicating the great number of very small changes, the weak shoulders, indicating fewer than normal moderate changes, and the big tails, indicating more than normal radical departures from the previous year's budget.

So far, every study examining public budgets has found a pattern similar to that shown in Figure 2.1. Jordan (2003) finds punctuated budget change distributions for US local expenditures; Robinson (2004), for Texas school districts; Breunig and Koski (2006), for state budgets; and Jones and Baumgartner (2005), for US national outlays since 1800. The pattern also emerges in other countries, including the United Kingdom (John and Margetts 2003), Denmark (Mortensen 2005), Germany (Breunig 2006), France (Baumgartner, François, and Foucault 2006), Belgium (Jones et al. 2009), Spain (Caamaño-Alegre and Lago-Peñas 2011; Chaqués-Bonafont, Palau, and Baumgartner 2015), and South Africa (Pauw 2007). Indeed, the results are so strong and invariant that punctuated equilibrium has been classified as “a general empirical law of public budgets” (Jones et al. 2009).

This suggests that we need a broader theory of how policy punctuations occur, one that is not so tightly tied to pluralistic forms of government. It is likely that different systems lead to different intensities in punctuations, yet don't escape the process—because it is rooted in the capacities of government to process information and allocate attention.

Work done by a team of researchers analyzing budget data from seven Western democracies showed that all the national-level frequency distributions not only could be characterized as leptokurtic but that the degree of leptokurtosis varied with the level of institutional friction (Jones et al. 2009). The higher the levels of friction in a nation's political system, the greater the extent of punctuations in the budget data. The stochastic process studies of public budgets indicate that friction does indeed lead to more

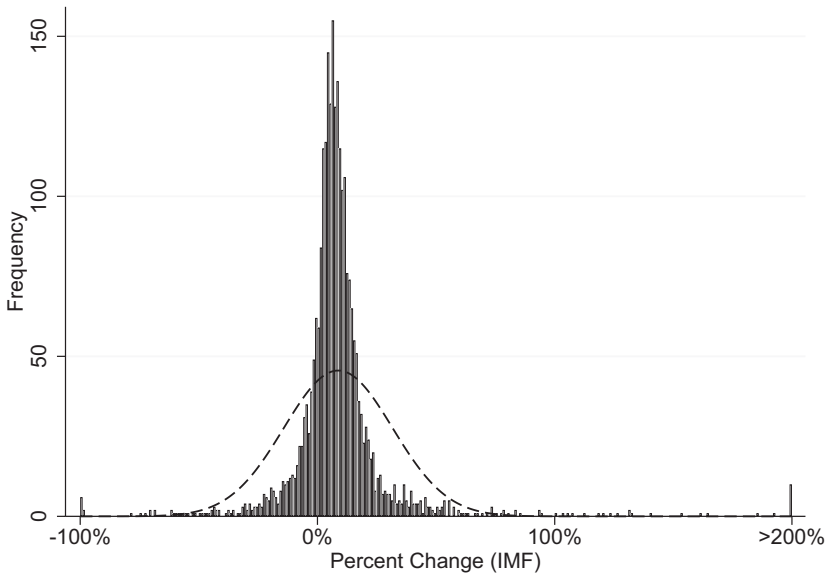


Figure 2.1 Annual Percentage Change in Ten Budget Categories in Twenty-Four OECD Countries, 1996–2011

Source: Created by the authors.

stability, as noted by political economy scholars such as Tsebelis (2002), but also to more dramatic changes than anticipated by those scholars.

Punctuated Equilibrium Theory in Comparative Perspective

The punctuated equilibrium model was originally developed to understand the dynamics of policy change in subsystems, but it has been extended to a more general formulation of punctuated change in policymaking, such as public budgeting. This has resulted in new insights into the process, including (1) an elaboration of an agenda-based, attention-driven budgeting model, (2) the generation of hypotheses concerning the distribution of annual budget changes and the reasons for its shape, and (3) empirical evidence that conforms to the new theory but is antithetical to the normal changes expected from incremental theory or from most other budget theories. Punctuated equilibrium, rather than incrementalism alone, characterizes national budgeting in America and elsewhere, just as punctuated equilibrium, rather than gridlock or marginalism, characterizes overall policymaking in the American political system.

But how general are these dynamics? Do they hold across political systems? The ubiquity of serial attentiveness and organizational routines of operation leads us to expect that stability and punctuations are a feature of policymaking in many governments. At the same time, the institutional

aspect of multiple venues interacts with boundedly rational decision-making to make Punctuated Equilibrium Theory particularly apt for relatively open democracies. It is likely that the processes of stability enforced by organizational routines interrupted by bursts of activity due to shifts in collective attention are general ones but that these processes are mediated by political institutions.

Where multiple venues occur as a consequence of institutional design, such as in federal systems, one would expect the dynamics of punctuated equilibrium to emerge. In the United States, overlapping jurisdictions of policy venues offer opportunities for issue entrepreneurs to change jurisdictions by emphasizing particular issue characterizations (Baumgartner, Jones, and McLeod 2000). To what extent does this kind of dynamic extend beyond US policymaking organizations? Sheingate (2000) has used the basic punctuated equilibrium to study agriculture policy in the European Union and the United States, and Pralle (2003) has studied environmental groups' exploitation of policy venues in forest policy in Canada and the United States. These systems have the requisite elements of openness and multiple venues. In the case of the European Union, the emergence of a strong central government from what previously were fully independent governments has offered students of public policy processes the opportunity to observe the effects of new venues for policy change. Princen and Rhinard (2006, 1) write that "agenda setting in the EU takes place in two ways: 'from above,' through high-level political institutions urging EU action, and 'from below,' through policy experts formulating specific proposals in low-level groups and working parties." Schrad (2010) used the idea to explain the global wave of prohibition in Western countries in the early twentieth century. Boushey (2010) applies the theory to how policies diffuse across the US states.

These interacting venues operate in many ways similarly to the pluralistic policymaking system in the United States (Wendon 1998; Guiraudon 2000; Mazey and Richardson 2001; Cichowski 2006). But such venue shopping does not always aid disadvantaged groups, as Guiraudon (2000) shows for immigration policy in France, Germany, the Netherlands, and the European Union, losing in a narrow venue does not mean winning in a broader one; it could instead invite even bigger losses (see also Mortensen 2007, 2009). When immigration rights organizations won victories in national courts, conservatives on the issue were able to appeal to the European Union and blunt their victories (see also Givens and Ludke 2004). Losers in one venue may also lose in the next.

If policymaking devolves to experts in all systems, then a key question is: when does the subsystem dominate, and when does the issue spill over into the broader macropolitical arena? Timmermans and Scholten (2006) suggest that, even in the technical arena of science policy in a smaller European parliamentary system—the Netherlands—this does occur, and again the dynamics are roughly similar to those highlighted in the American version of the punctuated equilibrium model. In a study of immigration policy, Scholten

and Timmermans (2004) show that immigration policy is punctuated but damped down through the implementation process at the local level.

Punctuated-type dynamics also occur in other European countries. Maesschalck (2002), in a study of a major police failure in Belgium in the Dutroux child abuse scandal, shows that policymaking generated by scandal follows a conflict expansion model consistent with the punctuated equilibrium approach. This finding is no fluke. In a comprehensive study of Belgian public policy processes during the 1990s, Walgrave, Varone, and Dumont (2006) directly compare the party model with the issue expansion model. They note the ability of the Dutroux and other scandals to destabilize the system, basically disrupting the party-dominated policymaking system with highly emotive information that political elites cannot afford to ignore. Similarly, John (2006a) finds that the interaction of media coverage and events is more important in explaining major changes in budget commitment for urban affairs in the United Kingdom than are changes in party control.

Cross-country studies of issue expansion offer the opportunity to examine how different institutional arrangements—that is, variations in the nature of political venues—affect the course of public policy. Timmermans (2001) examined cases of biomedical policy in four countries (Canada, the Netherlands, the United Kingdom, and Switzerland), finding that variation in arenas both at the macropolitical and policy subsystem levels had major effects on the tempo of agenda dynamics. Even where policy dynamics are broadly similar, as they seem to be in European democracies, the specific paths of policy development can be varied.

In this enterprise, we need the qualitative studies of Pralle (2003), Princen and Rhinard (2006), and Timmermans and Scholten (2006), as well as quantitative studies capable of tracing policy changes across longer periods (Cashore and Howlett 2007; Daviter 2009). Many of these studies focus on how institutional structures permit change to occur and how the institutional structure influences the speed and magnitude of policy dynamics (Mortensen 2005, 2007, 2009; Chaqués-Bonafont and Palau 2009).

The Comparative Policy Agendas Project

One of the major outcomes of the policy agenda-setting research initiated by Baumgartner and Jones in the early 1990s is the development of an international community of scholars doing policy agendas research. Within this loosely structured Comparative Agendas Project (CAP), some scholars apply and extend PET to countries other than the United States, but many scholars also work with other theories and other research questions. What unites these scholars is the application of the measurement system originally developed to construct the databases of the US-based Policy Agendas Project, as later adjusted for comparative use (Jones 2016).

Currently, the CAP consists of more than twenty country projects, with more in the pipeline. Every year in June the group of scholars organizes an annual meeting with around eighty participants on average.

Whereas the range of theoretical approaches and research questions is broad, a main advantage from a comparative perspective is the strict enforcement across countries of a common measurement system. This is a necessary requisite for further development of comparative research. Another recent initiative to promote more comparative policy agenda-setting research is the setup of a common webpage from which all country datasets can be downloaded and where students and researchers can easily conduct online analyses of the data. CAP's website, www.comparativeagendas.net, is hosted by the University of Texas. The Congressional Bills Project and the larger US-based Policy Agendas Project are all integrated into the cross-national Comparative Agendas Project.

To give an impression of the scope and direction of research inspired by either PET or the methodological approach to agenda setting affiliated with the theory, we performed a set of systematic keyword searches in the major online bibliographic databases Scopus and ProQuest. The search strings were "punctuated equilibrium Jones Baumgartner," "disproportionate information processing Jones Baumgartner," and "agenda setting Jones Baumgartner." Publications just mentioning or briefly referring to the PET or the measurement system of CAP were then excluded. In the second round of searching, we supplemented this list of publications with more ad hoc online searches of relevant websites like the CAP and the European Union Policy Agendas Project (<http://www.policyagendas.eu>). In the third round, we circulated the list to Comparative Agendas Project scholars and asked them to identify whether some of their relevant publications were missing from the list. As of April 1, 2022, we ended up with a total of 550 relevant publications covering the period from 1991 to the present. For the full list of publications, see the online resources for this chapter at www.routledge.com/9781032311296.

Figure 2.2 provides a sense of the growth in use of the theory and the measurement system over time. First, the figure inflects sharply upward after 2005. Second, there is a marked increase in the number of publications using non-US data. The number of US studies has also increased over the years, but a total of 65% of all empirical studies before 2006 were based solely on US data, whereas the equivalent number is reduced to 28% for studies published after 2005. The bulk of the studies now focus on political systems outside the United States. Third, the two most senior authors of this chapter were listed as coauthors for 23% of the publications in the early period, but for only 10% of those after 2005. Fourth, although the figure does not represent this finding, the literature is highly empirical: our review showed just twenty-six review articles (such as this one), forty-nine purely theoretical treatments, and 475 empirical works; this last category represents 86% of the publications in the field. Finally, a closer look at the list of publications shows a recent increase in publications in authoritarian and semi-authoritarian regimes. Thus, we can see that the CAP is developing into a truly comparative project, and not just a project for the usual comparisons of Western democracies.

Table 2.1 shows the journals in which these articles have appeared. This makes clear the importance of comparative work; 138 different journals are

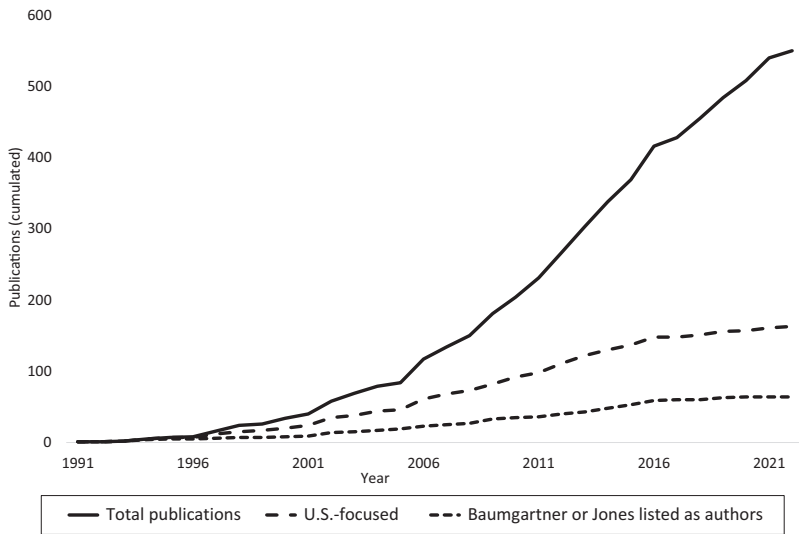


Figure 2.2 Punctuated Equilibrium Articles over Time

Source: Created by the authors.

represented in the list, including virtually all the major disciplinary journals as well as those focused on public policy, public administration, and US and comparative politics.

The dramatic increase in comparative use of the ideas and policy agenda-setting approach associated with the PET is a remarkable development given the initial response to what some perceived as a peculiarly American focus in Baumgartner and Jones's *Agendas and Instability in American Politics*, with its notions of venue shopping in the complicated US system of separation of powers, federalism, and weak political parties. Surely, some surmised, things must be different in more centralized systems with more disciplined political parties. Although many things are indeed different in each political system, the basic limits of human cognition, organizational capacity, and attention at the core of PET give the theory a potential for universal application.

The increase in the number of publications around 2006 coincides with the development of the network of the comparative policy agendas project formed around independent country projects. An overview of the country projects and databases can be found at www.comparativeagendas.net.

CAP offers the possibility of tracing policy dynamics in multiple countries simultaneously. In Denmark, Green-Pedersen and his collaborators have done so by studying tobacco policy in Denmark and the United States (Albaek, Green-Pedersen, and Nielsen 2007), euthanasia in Denmark, Belgium, and the Netherlands (Green-Pedersen 2004), and health care in Denmark and the United States (Green-Pedersen and Wilkerson 2006).

Another line of research where the issue attention approach has led to new important insights is the classic question about the interplay between

Table 2.1 Journals in Which Punctuated Equilibrium Articles Have Been Published

<i>Journal</i>	<i>Articles</i>
<i>Policy Studies Journal</i>	61
<i>Journal of European Public Policy</i>	26
<i>Journal of Public Policy</i>	24
<i>Policy Sciences</i>	23
<i>West European Politics</i>	13
<i>Party Politics</i>	13
<i>American Journal of Political Science</i>	11
<i>Journal of Public Administration Research and Theory</i>	10
<i>Review of Policy Research</i>	10
<i>Political Research Quarterly</i>	8
<i>Public Administration</i>	8
<i>Political Communication</i>	8
<i>European Journal of Political Research</i>	8
<i>Political Studies</i>	8
<i>Governance</i>	8
<i>Italian Political Science Review</i>	8
<i>Comparative Political Studies</i>	7
<i>European Political Science Review</i>	7
<i>Canadian Journal of Public Policy</i>	6
<i>British Journal of Political Science</i>	6
<i>Journal of Politics</i>	5
117 other journals, combined	166
Nonjournal publications (books, book chapters)	106
Total publications	550

Source: Created by the authors.

media and politics. In Belgium, Walgrave and colleagues have taken a lead on this research agenda (Walgrave 2008; Walgrave, Lefevere, and Nuytemans 2009; Vliegthart and Walgrave 2011), but the question has also been studied intensively in other countries such as Denmark (Green-Pedersen and Stubager 2010; Thesen 2013; Green-Pedersen, Mortensen, and Thesen 2015), the Netherlands and Switzerland (Sciarini, Tresch, and Vliegthart 2020), as well as across countries (Vliegthart et al. 2016). The mechanisms of issue expansion and policy development are broadly similar in different democratic political systems, even though they may play out differently as they are channeled through different decision-making institutions.

In an edited book (Engeli, Green-Pedersen, and Larsen 2012), scholars analyze morality politics through focused case comparisons of the treatment of such issues as euthanasia, abortion, and in vitro fertilization in the United States and several Western European countries. Green-Pedersen and Walgrave

(2014) have edited a book bringing together more quantitative findings from the Comparative Agendas Project in Europe to develop a theory not only of policy agendas and agenda setting but also of the dynamics of institutional evolution more generally. John et al. (2013) developed a similar argument about the development of UK politics from 1945 to present, and John and Jennings (2010) examined punctuated changes in governmental attention in the UK, finding that these punctuations were often unrelated to electoral turnover. Chaqués-Bonafont, Palau, and Baumgartner (2015) review recent Spanish political history through the lens of the Spanish agendas project and document a range of new insights from a country that quite recently transitioned to democracy.

Part of any differences in policies between countries may be attributed to differences in the mobilization of actors and the subsequent timing and sequencing of events. Consequently, even differences in policies between countries cannot necessarily be attributed to differences in institutions, as Pralle (2006) has shown in a case study of lawn pesticide policy in Canada and the United States. Jumping to the conclusion that Canada provides a more receptive venue for pesticide regulation might not be warranted without a study of the dynamics of political choice.

The punctuated equilibrium model is also proving useful in understanding relations among nations, such as in protracted interstate rivalries (Cioffi-Revilla 1998), the role of norms in international politics (Goertz 2003), and agenda setting in global disease control (Shiftman 2003; Shiftman, Beer, and Wu 2002). The latter study compared three models of policymaking—the incrementalist, the rationalist, and punctuated equilibrium—and found “a more complex pattern in which interventions are available only to select populations, punctuated with bursts of attention as these interventions spread across the globe in concentrated periods of time” (Shiftman, Beer, and Wu 2002, 225).

It is critical to begin to understand which aspects of policymaking result from more general dynamics based on human cognition and organizational behavior and which relate to the particulars of the institutions under study. Such considerations move us beyond the confines of theories for institutions and toward a more general theory of the interaction of humans in organizations (Goertz 2003).

The study of party competition from an issue attention perspective represents another new research area, focusing primarily on parliamentary systems. Two subthemes can be identified within this research agenda. One track offers a new approach to the classic question about the importance of elections and changes in the partisan composition of governments as drivers of policy and agenda changes. Research in this perspective challenges conventional wisdom about the importance of elections and ideology as explanations of change. Mortensen et al. (2011), for instance, in a study of change and stability in government agendas across three different countries and several decades, conclude that there is no evidence that elections, changes in government colors, or changes of prime minister systematically affect the

level of change and stability in government agendas. This finding corresponds with studies of agenda setting in France (Baumgartner, Grossman, and Brouard 2009), the United States (Jones and Baumgartner 2005, 84–85), and the United Kingdom (John and Jennings 2010) and across a large number of countries (Green-Pedersen and Walgrave 2014). Agendas do change over time, but the timing of such changes is not closely related to elections or shifts of governments, as Grossman and Guinaudeau (2021) report in a comprehensive study of issue agendas across five democracies since the 1980s.

The other subtheme within this research agenda on political parties regards how political parties compete with each other when trying to set the political agenda. Though most scholars acknowledge that parties do respond to their competitors' attention to issues, the dominant theoretical accounts (Budge and Farlie 1983; Petrocik 1996) have had much more to say about selective issue emphases (or "issue ownership") than about issue overlap and responsiveness. Agenda-setting studies challenge this conventional understanding by showing that political parties do respond to each other on the same issues instead of simply talking past each other on different issues (Vliegenthart and Walgrave 2011). Furthermore, the literature has focused on the unequal agenda-setting power of different political parties. Green-Pedersen and Mortensen (2010) provided a key insight into the roles of the parliamentary opposition in defining the political agenda in Denmark, discerning both the limits of strategic agenda setting available to the government and the power of the opposition to focus attention in those areas the government might prefer to avoid (see also Seeberg 2013; Green-Pedersen and Mortensen 2015; Green-Pedersen, Mortensen, and So 2018; Abou-Chadi, Green-Pedersen, and Mortensen 2020). In a recent book, Green-Pedersen (2019) advances the understanding of party issue competition in a comprehensive study of the issue agendas of seven West European countries since 1980.

The European Union Policy Agendas Project represents another important development in which a number of researchers have utilized the topics coding system to systematically trace attention to policy issues in the European Union (e.g., Princen 2009, 2013; Alexandrova and Timmermans 2013; Alexandrova 2015; Princen, Siderius, and Villasante 2021). Central questions within this research agenda include which issues appear on the EU agenda and how the definitions of issues change.

One promising new development is the application of the issue attention measurement system to local policy agendas. The local policy agenda is of crucial importance if we want to understand how local political systems work and function. Furthermore, the large number of comparable, local political systems offers new opportunities for systematic examination of the role of institutions, actors, and information in politics. A Danish research group led by Mortensen has been at the forefront of this new local policy agendas research (Baekgaard, Mortensen, and Seeberg 2018; Baekgaard, Larsen, and Mortensen 2019; Mortensen, Loftis, and Seeberg 2022), but projects are also underway in Norway and Austin.

Finally, vibrant work on PET is ongoing in several Asian countries (e.g., Hong and Sohn 2014; Yoon 2015; Chan and Zhao 2016; Yao, Yan, and Zhu 2021). The application of the ideas to new countries illustrates that the insights contained in Baumgartner and Jones's 1993 book were not limited to the time period, the political system, or the particular cases studied in that book. The perspective has proven to have much broader applicability, and we have no doubt that many of the insights and major findings will come from non-US locales.

Conclusion

The initial theory of punctuated equilibrium in policy processes is applicable to the dynamics of the specialized politics of policy subsystems. It has proved useful enough that scholars have employed it to understand a variety of policymaking situations in the United States and abroad. It has proved robust enough to survive several rigorous quantitative and qualitative tests. It has spawned a new approach to the study of public budgeting based on stochastic processes, and it hence has satisfied the criterion that any theory be not only falsifiable but also fruitful in suggesting new lines of inquiry.

It has also led to considerable discussion among policy practitioners. In his call to action on environmental change, *Red Sky at Morning*, Speth (2004) cites Punctuated Equilibrium Theory as a policy analysis that can lead to rapid, correcting change in the face of accumulating factual evidence. *Theories of the Policy Process* aims to supply better theory in the study of policy processes, and better applied work on policy change will occur with better theory; indeed, there is no substitute for this.

The formulation of the theory in stochastic process terms has made it possible to compare policy process theories with general formulations of human dynamic processes. Punctuated dynamics, in which any activity consists of long periods of stability interspersed with bursts of frenetic activity, may be the general case in human systems. For example, Barabasi (2005) shows that when humans prioritize incoming information for action, the distribution of waiting times for action on the information is "heavy tailed"—that is, leptokurtic. When prioritization is not practiced and inputs are instead subject to random choice for processing, the distribution is not fat tailed.

The complexity and changing interactions of the policy process mean that accurate policy predictions will be limited to the system level. Complex adaptive systems are characterized by nonlinearity, interdependencies, and complex interactions of variables across time. As a consequence, clear causal chains and precise predictions work only in some cases and during some times. Because stasis characterizes most of the cases and most of the times, scholars may be convinced that they have a good working model of the process. But a complete model will not be locally predictable because we cannot foresee the timing or the outcomes of the punctuations.

What will cause the next big shift in attention, change in dimension, or new frame of reference? Careful study of a policy or issue area may lead to

inferences about pressures for change, but when will the next attention shift occur in that particular policy area? At the systems level, punctuated equilibrium, as a theory, leads us to expect that some policy punctuation is under way almost all of the time. And the theory joins institutional settings and decision-making processes to predict that the magnitude of local changes will be related to their systems-level frequency of occurrence. Punctuated Equilibrium Theory will not help us make point-specific predictions for particular policy issues, unless we look only during periods of stability.

Linear predictions about the details of future policies will fail each time they meet an unforeseen punctuation; they will succeed only as long as the parameters of the test coincide with periods of equilibrium. This limitation means that it will be tempting to offer models applicable only to the more easily testable and confirmable periods of relative stability. Or investigators will focus on big changes and work backward from them to try to explain the case. This approach is subject to the fallacy of attributing causality to spurious factors. In our view, a clearer, more complete, and more empirically accurate theoretical lens is that of punctuated equilibrium, especially in its more general form, which integrates large policy changes with periods of stability.

Note

- 1 Special issues of the following journals have also appeared: *Policy Studies Journal* 41, no. 1 (2012); *Comparative Political Studies* 44, no. 8 (2011); and *Journal of European Public Policy* 13, no. 7 (2006).

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